C 40608 **(Pages : 2)** **Name.....**

Reg. No.....

SIXTH SEMESTER U.G. DEGREE EXAMINATION, MARCH 2023

(CBCSS–UG)

Mathematics

MTS 6B 14 (E01)—GRAPH THEORY

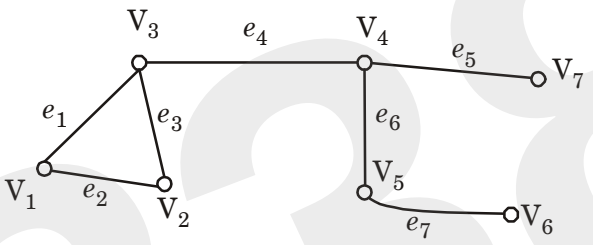
(2019 Admission onwards)

Time : Two Hours Maximum Marks : 60

Section A (Short Answer Questions)

Each question carries 2 *marks.*
A maximum of 20 *marks can be earned from this section.*

1. Find the number of edges of K_7.
2. Draw the graph $K_{3,3} - \{V\}$ where V is a Vertex in $K_{3,3}$.
3. Define K-regular graph. Give an example.
4. Define union of two graphs G_1 and G_2.
5. Let G be a simple graph with n vertices and $\overline{G}$ be its complement. Prove that, for each vertex V in G, $d_G(V) + d_{\overline{G}}(V) = n - 1$.
6. Define the adjacency matrix of a graph G with n vertices.
7. A connected graph G has 17 edges, what is the maximum possible number of vertices in G ?
8. Let



Find all bridges in G.

Turn over

9. When can you say that the complete graph $K_n, n \geq 3$ is Euler ? Justify.
10. Define critical planar graphs. Which complete graph K_n are critical planar ?
11. State Cayley's theorem on spanning trees.
12. When can you say that a graph G is maximal non-Hamiltonian.

(Ceiling marks = 20 marks)

Section B (Paragraph/Problem Type Questions)

Each question carries 5 *marks.*
A maximum of 30 *marks can be earned from this section.*

13. Prove that the complete bipartite graph $K_{3,3}$ is non-planar.
14. Let G be a simple planar graph with less than 12 vertices. Prove that G has a vertex V with $d(v) \leq 4$.
15. Prove that, in any graph G there is an even number of odd vertices.
16. Prove that for any connected graph G, rad G $\leq$ diam G $\leq$ 2. rad G.
17. Let u and v be distinct vertices of a tree 7. then prove that theme is precisely one path from u to v.
18. Prove that a simple graph G is Hamiltonian if and only if its closure c(G) is Hamiltonian.
19. Let G be a graph in which the degree of every vertex is at least two. Then prove that G contains a cycle.

(Ceiling marks = 30 marks)

Section C (Essay Type Questions)

Answer any **one** *question.*
The question carries 10 *marks.*

20. Explain the Konigsberg bridge problem. Give the graph theory model for this problem. Also state the respective theorem to solve this problem.
21. Let G be a graph with n vertices. Then prove that the following statements are equivalent :
    (i) G is a tree.
    (ii) G is acyclic graph with $n - 1$ edges.
    (iii) G is a connected graph with $n - 1$ edges.

(1 × 10 = 10 marks)

C 40608-A **(Pages : 4)** **Name.....**

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SIXTH SEMESTER U.G. DEGREE EXAMINATION, MARCH 2023

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Mathematics

MTS 6B 14 (E01)—GRAPH THEORY

(2019 Admission onwards)

(Multiple Choice Questions for SDE Candidates)

Time : 15 Minutes	**Total No. of Questions : 15**	**Maximum : 15 Marks**

INSTRUCTIONS TO THE CANDIDATE

1. This Question Paper carries Multiple Choice Questions from 1 to 15.
2. The candidate should check that the question paper supplied to him/her contains all the 15 questions in serial order.
3. Each question is provided with choices (A), (B), (C) and (D) having one correct answer. Choose the correct answer and enter it in the main answer-book.
4. The MCQ question paper will be supplied after the completion of the descriptive examination.

MTS 6B 14 (E01)—GRAPH THEORY

(Multiple Choice Questions for SDE Candidates)

1. What is the maximum number of edges in a bipartite graph having 12 vertices ?

   (A) 24. (B) 38.

   (C) 36. (D) 32.

2. A connected planar graph having 6 vertices, 7 edges contains ———————— regions.

   (A) 1. (B) 3.

   (C) 5. (D) 7.

3. Which of the following statements is/are TRUE for undirected graphs ?

   P : Number of odd degree vertices is even.

   Q : Sum of degrees of all vertices is even.

   (A) P only. (B) Q only.

   (C) Both P and Q. (D) Neither P and Q.

4. Which of the following statements is/are TRUE for Tree ?

   P : Every two points of G are joined by a unique path.

   Q : G is connected and $p = q - 1$.

   (A) P only. (B) Q only.

   (C) Both P and Q. (D) Neither P and Q.

5. What is the radius of the Petersen graph ?

   (A) 2. (B) 3.

   (C) 4. (D) None of the above.

6. Let G = P_n . Then :

   (A) Every sub graph of G is path.

   (B) Every proper sub graph of G is path.

   (C) Every spanning sub graph of G is path.

   (D) Every induced sub graph of G is path

7. Which of the following statement is/are TRUE ?

P : A cycle is a walk with end vertices are same.

Q : A cycle is a path with end vertices are same.

(A) P only. (B) Q only.

(C) Both P and Q. (D) Neither P and Q.

8. Number of edges incident with the vertex V is called ?

(A) Degree of a graph. (B) Handshaking lemma.

(C) Degree of a vertex. (D) None of the above.

9. If a simple graph G. contains n vertices and m edges, the number of edges in the Graph G′ (Complement of G) is ————————.

(A) $\frac{n^2 + n - 2m}{2}$. (B) $\frac{n^2 + 2n - 2m}{2}$.

(C) $\frac{n^2 - n - 2m}{2}$. (D) $\frac{n^2 - 2n - 2m}{2}$.

10. The number of elements in the adjacency matrix of a graph having 7 vertices is :

(A) 7. (B) 14.

(C) 36. (D) 49.

11. If G is the forest with 54 vertices and 17 connected components, G has ———————— total number of edges.

(A) 37. (B) 71.

(C) 17. (D) 54.

12. What is the maximum number of possible non zero values in an adjacency matrix of a simple graph with n vertices ?

(A) n^2. (B) $\frac{n(v-1)}{2}$.

(C) $\frac{n(n+1)}{2}$. (D) n.

Turn over

13. Radius of a graph, denoted by rad(G) is defined by ———————— ?

(A) $\text{Max}\{e(v) : v \in V(G)\}$.

(B) $\text{Min}\{e(v) : v \in V(G)\}$.

(C) $\text{Max}\{d(v, v) : v, u \in V(G)\}$.

(D) $\text{Min}\{d(u, v) : v, u \in V(G)\}$.

14. In a 2-connected graph G, any two longest cycles have at least ———————— vertices in common.

(A) 0. (B) 1.

(C) 2. (D) 3.

15. Let G be a bipartite graph.

P : Any vertex deleted graph G – v is also a bipartite graph.

Q : There exist two disjoint trivial induced subgraphs of G.

(A) P is true and Q is false. (B) P is false and Q is true.

(C) P and Q are false. (D) P and Q are true.